

31. Ethernet Common Agent (COMA)

31.1 Overview

Ethernet Common Agent (COMA) is an agent of ESWM which aims at gathering common functionalities for other agents. COMA handles the APB to SFR bus conversion, the buffer pointer release for rejected frames and the pointer handling for local RAM. Software is required for the configuration of Ethernet Common Agent.

Table 31.1 lists the COMA specifications and Figure 31.1 shows the block diagram.

Table 31.1 COMA specifications

Item	Function	Description
Overall function	Pointer management	Manage the local RAM pointers and their distribution
	APB to SFR	- Convert APB bus to SFR bus - Distribute SFR access among other switch IPs
	Descriptor reject	Reject the unused descriptors by agents for transmission
Data provision	Fabric error bus	Pointer received from fabric through error bus
	Descriptor bus	Descriptor received from MFWD for rejection
Control function	RAM access	Function that writes and reads to RAM

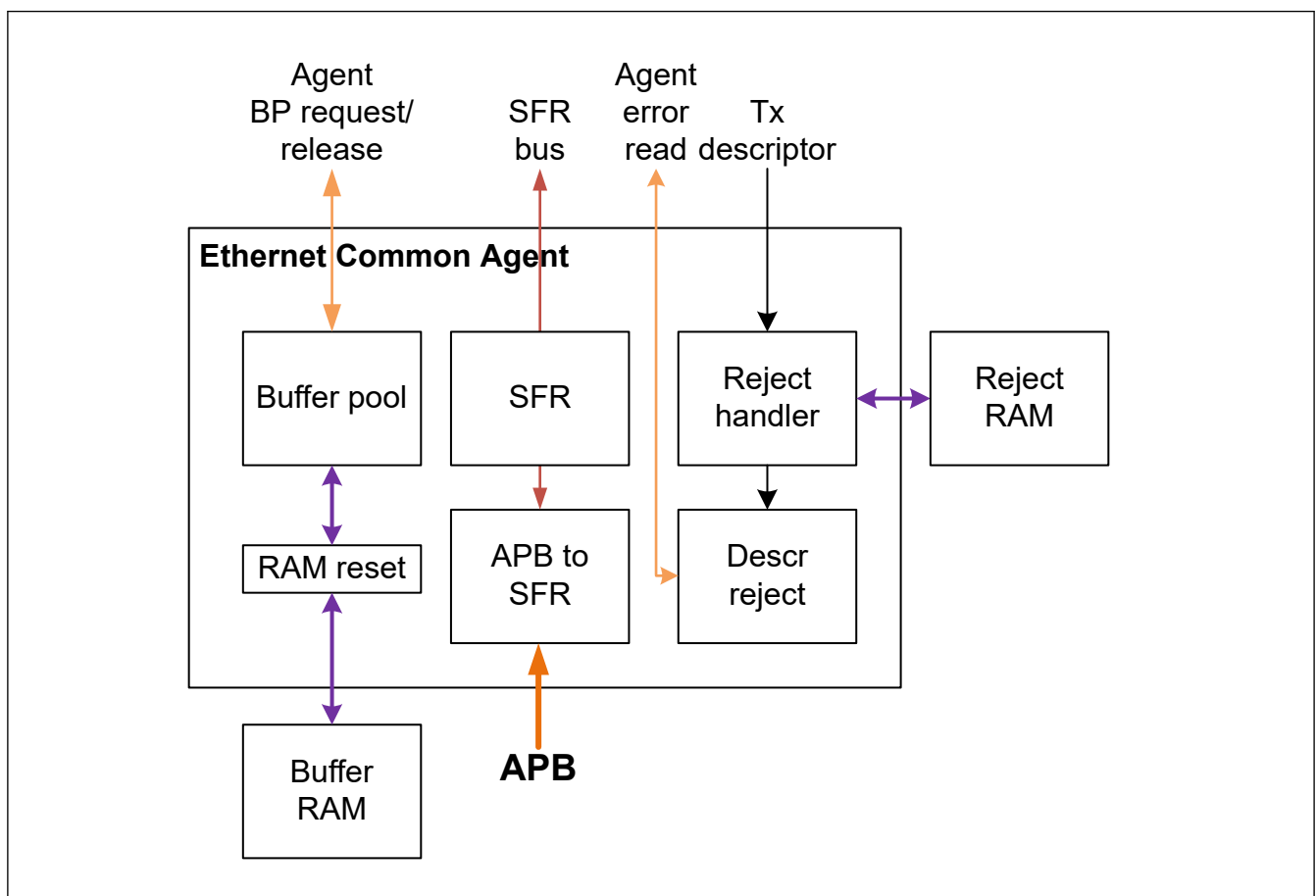


Figure 31.1 **COMA block diagram**

Table 31.2 COMA functional blocks (1 of 2)

Block name	Function
APB to SFR	Convert APB bus to SFR bus for every IP in ESWM
SFR	Ethernet Common Agent SFR

Table 31.2 COMA functional blocks (2 of 2)

Block name	Function
Reject handler	Receive descriptors to be rejected from forwarding engine and save them in reject RAM
Descriptor reject	Reject the descriptors
Free buffer pool	Handles the free buffer pointers
RAM reset	Handle RAM reset

31.2 COMA Register List

Table 31.3 COMA registers

Offset address	Register symbol	Register name
0x0000	RIPV	IP Version Register
0x0004	RRC	Reset Configuration Register
0x0008	RCEC	Clock Enable Configuration Register
0x000C	RCDC	Clock Disable Configuration Register
0x0020 + 0x04 × i	CABPIBWMCi	Buffer Pool IPV Based Watermark Configuration Register i (i = 0 to 7)
0x0040	CABPWMLC	Buffer Pool Watermark Level Configuration Register
0x0050 + 0x04 × i	CABPPFLCi	Buffer Pointer Pause Frame Level i Configuration Register (i = 0, 1)
0x0060 + 0x04 × i	CABPPWMLCi	Port i Buffer Pool Watermark Level Configuration Register i (i = 0 to 2)
0x00A0 + 0x04 × (i × 2 + j)	CABPPPFLCij	Port i Buffer Pointer Pause Frame Level j Configuration Register (i = 0 to 2) (j = 0, 1)
0x0100 + 0x04 × i	CABPULCi	Port i Buffer Pointer Utilization Level Configuration Register i (i = 0 to 2)
0x0140	CABPIRM	Buffer Pool Initialization Register Monitoring Register
0x0144	CABPPCM	Buffer Pool Pointer Count Monitoring Register
0x0148 E: 0x014C	CABPLCM	Buffer Pool Pointer Least Count Monitoring Register
0x0180 + 0x04 × i	CABPCPMi	Port i Buffer Pointer Count Monitoring Register (i = 0 to 2)
0x0200 + 0x04 × i E: 0x0280 + 0x04 × i	CABPMCPMi	Port i Buffer Pointer Maximum Count Monitoring Register (i = 0 to 2)
0x0300	CARDNM	Rejected Descriptor Number Monitoring Register
0x0304 E: 0x0308	CARDMNM	Rejected Descriptor Maximum Number Monitoring Register
0x0310 E: 0x0314	CARDCN	Rejected Descriptor Counter Register
0x0400	CAEIS0	Error Interrupt Status Register 0
0x0404	CAEIE0	Error Interrupt Enable Register 0
0x0408	CAEID0	Error Interrupt Disable Register 0
0x0410	CAEIS1	Error Interrupt Status Register 1
0x0414	CAEIE1	Error Interrupt Enable Register 1
0x0418	CAEID1	Error Interrupt Disable Register 1
0x0440	CAMIS0	Monitoring Interrupt Status Register 0
0x0444	CAMIE0	Monitoring Interrupt Enable Register 0
0x0448	CAMID0	Monitoring Interrupt Disable Register 0
0x0450	CAMIS1	Monitoring Interrupt Status Register 1
0x0454	CAMIE1	Monitoring Interrupt Enable Register 1
0x0458	CAMID1	Monitoring Interrupt Disable Register 1

Note: Base address: COMA = 0x403C_90000, COMA_NS = 0x503C_90000

Note: The address preceded by E: corresponds to an emulation address which allows a register to be read without modifying its content.

31.3 COMA Register Descriptions

31.3.1 ESWM Registers

31.3.1.1 RIPV : IP Version Register

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0000

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	CAIPV[3:0]			FBIPV[3:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	1	1	0	0	1	1
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	EAIPV[3:0]				FWIPV[3:0]				GWIPV[3:0]				TIPV[3:0]			
Value after reset:	0	0	1	1	0	0	1	1	0	0	1	1	0	0	1	1

Bit	Symbol	Function	R/W
3:0	TIPV[3:0]	Top Module IP Version Number	R
7:4	GWIPV[3:0]	Gateway CPU Agent IP Version Number	R
11:8	FWIPV[3:0]	Forwarding Engine IP Version Number	R
15:12	EAIPV[3:0]	Ethernet Agent IP Version Number	R
19:16	FBIPV[3:0]	Fabric Bus IP Version Number	R
23:20	CAIPV[3:0]	Ethernet Common Agent IP Version Number	R
31:24	—	These bits are read as 0.	R

31.3.1.2 RRC : Reset Configuration Register

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0004

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RR
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	RR	Software Reset 0: ESWM not reset 1: ESWM reset	R/W
31:1	—	These bits are read as 0. The write value should be 0.	R/W

RR bit (Software Reset)

- When set, this bit resets all the ESWM except the APB interface

- The synchronous reset RCC.RR is power-consuming, so it is only internally asserted for 7 clock cycles
- The synchronous reset is also internally asserted for 7 clock cycles after a hardware reset release.

[Caution]

- Because the synchronous reset RCC.RR is power-consuming, it should always be deasserted right after assertion to avoid power consumption.

31.3.1.3 RCEC : Clock Enable Configuration Register

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0008

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RCE
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ACE2 to ACE0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	ACE2 to ACE0	Agent i Clock Enable (i = 0 to 2) Bit i enables agent i clock. 0: Agent i clock disabled 1: Agent i clock enabled	R/W ^{*1}
15:3	—	These bits are read as 0. The write value should be 0.	R/W
16	RCE	Clock Enable 0: ESWM clock disabled 1: ESWM clock enabled	R/W ^{*1}
31:17	—	These bits are read as 0. The write value should be 0.	R/W

Note: ESWM has three system clocks:

- Peripheral clock: Can be stopped only by the upper layer (MCU)
- MFWD clock: Controlled by RCE
- Agent clock: Controlled by ACEi.

Note 1. Read value differs from written value.

The COMA registers are controlled by peripheral clocks, which can only be stopped by the upper layer (MCU). Therefore, COMA registers always have read and write access.

When peripheral clock is running and agent clock is stopped (ACEi is 0):

- GWCA and ETHA have APB read access but do not have APB write access.

When peripheral clock is running and MFWD clock is stopped (RCE is 0):

- MFWD has APB read access but does not have APB write access
- Agent clocks are stopped.

ACEi bits (Agent i Clock Enable (i = 0 to 2))

- Bit i enables agent i clock. For agent mapping details, see the Fabric specifications [MFAB].
- When RRC.RR register is set, all agent i clock is internally enabled to allow software reset even if RCC.ACEi is not set.
- When RCC.RCE register is not set, agent i clock is internally disabled even if RCC.ACEi is set.

[Setting condition]

- SW: Writing 1 to this bit.

[Clearing condition]

- SW: Writing 1 to RCDC.ACDi register.

RCE bit (Clock Enable)

The RCE bit enables the switch clocks. When RRC.RR register is set, ESWM clock is internally enabled to allow software reset even if RCC.RCE is not set.

[Setting condition]

- SW: Writing 1 to this bit.

[Clearing condition]

- SW: Writing 1 to RCDC.RCD register.

31.3.1.4 RCDC : Clock Disable Configuration Register

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x000C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	RCD
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	ACD2 to ACD0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	ACD2 to ACD0	Agent i Clock Disable (i = 0 to 2) 0: No effect 1: Clears bit i in RCEC.ACEi register	W*2
15:3	—	These bits are read as 0.	R
16	RCD	Clock Disable*1 Writing 1 to this bit clears RCEC.RCE register.	W*2
31:17	—	These bits are read as 0.	R

Note 1. SW: Only write to this bit when all agents are disabled and when CABPPCM.RPC is equal to 512.

Note 2. Read value differs from written value.

31.3.2 Ethernet Common Agent Function Registers

31.3.2.1 CABPIBWMCi : Buffer Pool IPV Based Watermark Configuration Register i (i = 0 to 7)

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0020 + 0x04 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	IBSWMPN[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	IBUWMPN[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	IBUWMPN[9:0]	IPV Based Unsecure Watermark Pointer Number ^{*1}	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
25:16	IBSWMPN[9:0]	IPV Based Secure Watermark Pointer Number ^{*2}	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Do not set this register to a value greater than the following:

- SW: 512
- SW: CABPIBWMC{i + 1}.IBUWMPN{i + 1}.

Note 2. Do not set this register to a value greater than the following:

- SW: 512
- SW: CABPIBWMC{i + 1}.IBSWMPN{i + 1}.

IBUWMPN[9:0] bits (IPV Based Unsecure Watermark Pointer Number)

- For i equal to 7, setting this register to 512 disables the IPV based watermark function for IPV 7
- For i other than 7, setting this register to CABPIBWMC{i + 1}.IBUWMPN{i + 1} disables the IPV-based watermark function for IPV
- See [section 31.5.1.1. IPV-based Watermark](#) for details.

IBSWMPN[9:0] bits (IPV Based Secure Watermark Pointer Number)

- For i equal to 7, setting this register to 512 disables the IPV based watermark function for IPV 7
- For i other than 7, setting this register to CABPIBWMC{i + 1}.IBSWMPN{i + 1} disables the IPV-based watermark function for IPV i
- See [section 31.5.1.1. IPV-based Watermark](#) for details.

31.3.2.2 CABPWMLC : Buffer Pool Watermark Level Configuration Register

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0040

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	WMCL[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	WMFL[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	WMFL[9:0]	Watermark Flush Level*1 - Setting this register to 512 disables the watermark flush level function - See section 31.5.1.2. Global Level-based Watermark for details	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
25:16	WMCL[9:0]	Watermark Critical Level*2 - Setting this register to 512 disables the watermark critical level function - See section 31.5.1.2. Global Level-based Watermark for details	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. SW: Do not set this register to a value greater than 512.

Note 2. SW: Always set this register to a value less than or equal to CABPWMLC.WMFL but not greater than 512.

31.3.2.3 CABPPFLCi : Buffer Pointer Pause Frame Level i Configuration Register

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0050 + 0x0004 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	PAL[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PDL[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	PDL[9:0]	Pause De-Assertion Level*1 See section 31.5.2.1. Global Pause Function for details.	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
25:16	PAL[9:0]	Pause Assertion Level*2 - Setting this register to 512 disables pause frames - See section 31.5.2.1. Global Pause Function for details.	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

Note 1.

- SW: Always set this register to a value less than or equal to CABPPFLCi.PAL.

- SW: Do not set this register to a value greater than 512 or less than 24.

Note 2. SW: Do not set this register to a value greater than 512 or less than 24.

PDL[9:0] bits (Pause De-Assertion Level)**[Precautions]**

If CABPPFLCi.PAL value is too close to CABPPFLCi.PDL value, the corresponding PAUSE signals change state rapidly. Additionally, if no automatic pause frames are activated [RMAC], this can result in a succession of pause frames on RMAC transmission which can decrease the link performance.

31.3.2.4 CABPPWMLCi : Port i Buffer Pool Watermark Level Configuration Register (i = 0 to 2)

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0060 + 0x0004 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	PWMCL[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PWMFL[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	PWMFL[9:0]	Watermark Flush Level* ¹ - Setting this register to 512 disables the watermark flush level function - See section 31.5.1.3. Per-port Level-based Watermark i (i = 0 to 3) for details	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
25:16	PWMCL[9:0]	Watermark Critical Level* ² - Setting this register to 512 disables the watermark critical level function - See section 31.5.1.3. Per-port Level-based Watermark i (i = 0 to 3) for details	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. SW: Do not set this register to a value greater than 512.

Note 2. SW: Always set this register to a value less than or equal to CABPPWMLCi.PWMFL but not greater than 512.

31.3.2.5 CABPPFLCij : Port i Buffer Pointer Pause Frame Level j Configuration Register (i = 0 to 2) (j = 0, 1)

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x00A0 + 0x04 × (i × 2 + j)

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	PPAL[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	PPDL[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	PPDL[9:0]	Pause De-Assertion Level* ¹ See section 31.5.2.2. Per-port Pause Function i (i = 0 to 3) for details. If CABPPPFLCij.PPAL value is too close to CABPPPFLCij.PPDLij value, the corresponding PAUSE signals change state rapidly. Additionally, if no automatic pause frames are activated [RMAC], this can result in a succession of pause frames on RMAC transmission which can decrease the link performance.	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
25:16	PPAL[9:0]	Pause Assertion Level* ² Setting this register to 512 disables pause frames. See section 31.5.2.2. Per-port Pause Function i (i = 0 to 3) for details.	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Restrictions:

- SW: Always set this register to a value less than or equal to CABPPPFLCij.PPAL.
- SW: Do not set this register to a value greater than 512 or less than 6.

Note 2. SW: Do not set this register to a value greater than 512 or less than 6.

31.3.2.6 CABPULCi : Port i Buffer Pointer Utilization Level Configuration Register (i = 0 to 2)

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0100 + 0x0004 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	MNNPN[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	MXNPN[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	MXNPN[9:0]	Maximum Number of Pointer* ¹ Maximum number of pointers that can be used by port i.	R/W
15:10	—	These bits are read as 0. The write value should be 0.	R/W
25:16	MNNPN[9:0]	Minimum Number of Pointer* ² Minimum number of pointers that can be used by port i.	R/W
31:26	—	These bits are read as 0. The write value should be 0.	R/W

Note: If CABPULCi.MNNPN[i] of port[i] is changed (reconfigured) from its default value, then it restricts the reception maximum frame size of ports other than port[i]. The restricted reception maximum frame size of ports other than port[i] in such case is given in the following expression.

Restricted reception maximum frame size = $((512 - \text{CABPULCi.MNNPN[i]}) / (3)) \times 128$ bytes. This is because ports other than port[i] must share the remaining pointers $(512 - \text{CABPULCi.MNNPN[i]})$.

If this restriction is not followed, then the consumed buffer pointers are not released by other ports and the frame is stuck inside the switch. Because the total buffer pointers are insufficient for the port in the middle of reception, to resolve this, reconfigure CABPULCi.MNNPN[i] value to a smaller value.

Note 1. SW: Set the maximum number of pointers to a value large enough to save two frames with a size equal to RMAC reception maximum frame size (set in RMAC).

Note 2. • SW: The sum of all CABPULCi.MNNPN should be smaller or equal to 512.
• SW: This register should always be smaller or equal to CABPULCi.MXNPN.

MXNPN[9:0] bits (Maximum Number of Pointer)

- If register CABPCPMi.RPCP is equal to this register value, port i no longer receives pointer
- See [section 31.5.3. Per-port Memory Allocation Function](#) for details.

MNNPN[9:0] bits (Minimum Number of Pointer)

- If $SUM(CABPULCi.MNNPN) \geq CABPPCM.RPC + SUM(CABPCPMi.RPCP)$ only ports for which $CABPCPMi.RPCP < CABPULCi.MNNPN$ can receive pointers
- See [section 31.5.3. Per-port Memory Allocation Function](#) for details.

31.3.2.7 CABPIRM : Buffer Pool Initialization Register Monitoring Register

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0140

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	BPR	BPIOG
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	BPIOG	Buffer Pool Initialization Ongoing ^{*1}	R/W ^{*2}
1	BPR	Buffer Pool Ready	R/W
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. HW: CPU cannot write 1 to this bit when CABPIRM.BPR is set.

Note 2. Read value differs from written value.

BPIOG bit (Buffer Pool Initialization Ongoing)

[Setting condition]

- SW: Writing 1 to this register.

[Clearing condition]

- HW: When buffer pool initialization is complete.

BPR bit (Buffer Pool Ready)

[Setting conditions]

- When CABPIRM.BPIOG is cleared
- HW: This bit is set at $\text{clk_period [ns]} \times 512$ time from the start of buffer pool initialization.

31.3.2.8 CABPPCM : Buffer Pool Pointer Count Monitoring Register

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0144

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	TPC[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	RPC[9:0]									
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
9:0	RPC[9:0]	Remaining Pointer Count	R
15:10	—	These bits are read as 0.	R
25:16	TPC[9:0]	Total Pointer Count	R
31:26	—	These bits are read as 0.	R

RPC[9:0] bits (Remaining Pointer Count)

[Increment condition]

- HW: Incremented by 1 when a pointer is released.

[Decrement condition]

- HW: Decrement by 1 when a pointer is given to an agent.

TPC[9:0] bits (Total Pointer Count)

- Read 512, fixed to 512
- Reading this register gives the number of pointers contained in the buffer pool. The switch local data RAM size is equal to $128 \times \text{CABPPCM.TPC}$ bytes.

31.3.2.9 CABPLCM : Buffer Pool Pointer Least Count Monitoring Register

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0148

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	LRC[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
12:0	LRC[12:0]	Least Remaining Pointer Count This register remembers the smaller value that is taken by the remaining counter pointer. [Update condition] - HW: This register is updated to CABPPCM.RPC when CABPPCM.RPC < CABPLCM.LRC. [Clear condition] - Reading this register clears it to CABPPCM.RPC value.	R
31:13	—	These bits are read as 0.	R

31.3.2.10 CABPCPMi : Port i Buffer Pointer Count Monitoring Register (i = 0 to 2)

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0180 + 0x0004 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	RPCP[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
12:0	RPCP[12:0]	Received Pointer Count	R
31:13	—	These bits are read as 0.	R

RPCP[12:0] bits (Received Pointer Count)

This register shows the number of pointers that are used by agent i to store data.

[Increment condition]

- HW: Incremented by 1 when a pointer is given to port I.

[Decrement condition]

- HW: Decrement by 1 when a pointer given to port i is being released.

31.3.2.11 CABPMCPMi : Port i Buffer Pointer Maximum Count Monitoring Register (i = 0 to 2)

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0200 + 0x0004 × i

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	RPMCP[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
12:0	RPMCP[12:0]	Received Pointer Maximum Count ^{*1}	R
31:13	—	These bits are read as 0.	R

Note 1. May exceed several upper and lower limit settings.

RPMCP[12:0] bits (Received Pointer Maximum Count)

This register remembers the biggest value that is taken by CABPMCPMi.RPCP since the last read access.

[Update condition]

- HW: This register is updated to CABPMCPMi.RPCP when $CABPMCPMi.RPCP > CABPMCPMi.RPMCP$.

[Clear condition]

- Reading this register clears it to CABPMCPMi.RPCP value.

31.3.2.12 CARDNM : Rejected Descriptor Number Monitoring Register

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0300

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	RDNR[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
12:0	RDNR[12:0]	Rejected Descriptor Number in Reject RAM	R
31:13	—	These bits are read as 0.	R

RDNR[12:0] bits (Rejected Descriptor Number in Reject RAM)

This register shows the number of descriptors currently contained in the reject RAM.

[Increment condition]

- HW: Incremented by 1 when a descriptor is received from the forwarding engine.

[Decrement condition]

- HW: Decrement by 1 when a descriptor is read from the reject RAM.

31.3.2.13 CARDNMN : Rejected Descriptor Maximum Number Monitoring Register

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0304

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	RDMNRR[12:0]												
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
12:0	RDMNRR[12:0]	Rejected Descriptor Maximum Number in Reject RAM	R
31:13	—	These bits are read as 0.	R

RDMNRR[12:0] bits (Rejected Descriptor Maximum Number in Reject RAM)

This register remembers the biggest value that is taken by CARDNM.RDNRR since the last read access.

[Update condition]

- Set to CARDNM.RDNRR when $CARDNM.RDNRR > CARDNMN.RDMNRR$.

[Clearing condition]

- Reading this register clears it to CARDNM.RDNRR value.

31.3.3 Ethernet Common Agent Counter Registers

This section describes Ethernet Common Agent Counter registers.

31.3.3.1 CARDCN : Rejected Descriptor Counter Register

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0310

Bit position:	31	0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
Bit field:	RDN[31:0]																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																														
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
31:0	RDN[31:0]	Rejected Descriptor Number	R

RDN[31:0] bits (Rejected Descriptor Number)

This register counts the number of descriptors that is rejected since the last read.

[Clearing condition]

- SW: Reading this register clears it.

[Increment condition]

- HW: Incremented by 1 when a rejected descriptor is received from MFWD and if this register has a value other than 0xFFFF.

31.3.4 Ethernet Common Agent Interrupt Registers

This section describes Ethernet Common Agent Interrupt registers.

31.3.4.1 CAEIS0 : Error Interrupt Status Register 0

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0400

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	WMFL OS	WMCL OS	BPOP S	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0. The write value should be 0.	R/W
8	BPOPS	Buffer Pool Out of Pointer Status Flag	R/W ^{*1}
9	WMCLOS	Watermark Critical Level Overtook Status Flag	R/W ^{*1}
10	WMFLOS	Watermark Flush Level Overtook Status Flag	R/W ^{*1}
31:11	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

BPOPS bit (Buffer Pool Out of Pointer Status Flag)

[Setting condition]

- HW: When CABPPCM.RPC becomes null.

[Clearing condition]

- SW: Writing 1 to this bit.

[Error recovery]

- SW: When there is no error recovery, the switch can overflow [RMAC]. When using Cut through, use the watermark function so that this status does not occur. If this error occurs (it may get hung-up in the transmit state [ETHA and RMAC]), initialize the state with the emergency reset flow [TOP].

WMCLOS bit (Watermark Critical Level Overtook Status Flag)

[Setting condition]

- HW: When watermark critical level WM.CRITICAL is asserted to all 1s because of global level-based watermark (see [Figure 31.12](#)).

[Clearing condition]

- SW: Writing 1 to this bit.

[Error recovery]

- SW: When there is no recovery, data can be lost [FWD].

WMFLOS bit (Watermark Flush Level Overtook Status Flag)

[Setting condition]

- HW: When watermark flush level WM.FLUSH is asserted to all 1s because of global level-based watermark (see [Figure 31.12](#)).

[Clearing condition]

- SW: Writing 1 to this bit.

[Error recovery]

- SW: When there is no recovery, data can be lost [FWD].

31.3.4.2 CAEIE0 : Error Interrupt Enable Register 0

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0404

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	WMFL OE	WMCL OE	BPOP E	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0. The write value should be 0.	R/W
8	BPOPE	Buffer Pool Out of Pointer Enable	R/W ^{*1}
9	WMCLOE	Watermark Critical Level Overtok Enable	R/W ^{*1}
10	WMFLOE	Watermark Flush Level Overtok Enable	R/W ^{*1}
31:11	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

BPOPE bit (Buffer Pool Out of Pointer Enable)

[Setting condition]

- Writing 1 to this bit.

[Clearing condition]

- Writing 1 to CAEID0.BPOPD register.

WMCLOE bit (Watermark Critical Level Overtok Enable)

[Setting condition]

- Writing 1 to this bit.

[Clearing condition]

- Writing 1 to CAEID0.WMCLOD register.

WMFLOE bit (Watermark Flush Level Overtok Enable)

[Setting condition]

- Writing 1 to this bit.

[Clearing condition]

- Writing 1 to CAEID0.WMFLOD register.

31.3.4.3 CAEID0 : Error Interrupt Disable Register 0

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0408

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	WMFL OD	WMCL OD	BPOP D	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	—	These bits are read as 0.	R
8	BPOPD	Buffer Pool Out of Pointer Disable Writing 1 to this bit clears CAEIE0.BPOPE register.	R
9	WMCLOD	Watermark Critical Level Overtook Disable Writing 1 to this bit clears CAEIE0.WMCLOE register.	R
10	WMFLOD	Watermark Flush Level Overtook Disable Writing 1 to this bit clears CAEIE0.WMFLOE register.	R
31:11	—	These bits are read as 0.	R

31.3.4.4 CAEIS1 : Error Interrupt Status Register 1

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0410

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	PWMFLOS2 to PWMFLOS0		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	PWMCLOS2 to PWMCLOS0		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	PWMCLOS2 to PWMCLOS0	Port n Watermark Critical Level Overtook Status Flag (n = 0 to 2)	R/W ^{*1}
15:3	—	These bits are read as 0.	R/W
18:16	PWMFLOS2 to PWMFLOS0	Port n Watermark Flush Level Overtook Status Flag (n = 0 to 2)	R/W ^{*1}
31:19	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

PWMCLOS_n bits (Port n Watermark Critical Level Overtook Status Flag (n = 0 to 2))

[Setting condition]

- HW: For bit i, when watermark critical level bit i WM.CRITICAL[i] is asserted to 1 because of per-port level-based watermark (see [Figure 31.13](#)).

[Clearing condition]

- SW: Writing 1 to one of these bits.

[Error recovery]

- When there is no error recovery, data can be lost [FWD].

PWMFLOSn bits (Port n Watermark Flush Level Overtook Status Flag (n = 0 to 2))

[Setting condition]

- HW: For bit i, when watermark flush level bit i WM.FLUSH[i] is asserted to 1 because of per-port level-based watermark (see [Figure 31.13](#)).

[Clearing condition]

- SW: Writing 1 to one of these bits.

[Error recovery]

- When there is no error recovery, data can be lost [FWD].

31.3.4.5 CAEIE1 : Error Interrupt Enable Register 1

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0414

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	PWMFLOE2 to PWMFLOE0		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	PWMCLOE2 to PWMCLOE0		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	PWMCLOE2 to PWMCLOE0	Port n Watermark Critical Level Overtook Enable (n = 0 to 2)	R/W ^{*1}
15:3	—	These bits are read as 0. The write value should be 0.	R/W
18:16	PWMFLOE2 to PWMFLOE0	Port n Watermark Flush Level Overtook Enable (n = 0 to 2)	R/W ^{*1}
31:19	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

PWMCLOEn bits (Port n Watermark Critical Level Overtook Enable (n = 0 to 2))

[Setting condition]

- Writing 1 to one of these bits.

[Clearing condition]

- SW: Writing 1 to one of CAEID1.PWMCLOD bits.

PWMFLOEn bits (Port n Watermark Flush Level Overtook Enable (n = 0 to 2))

[Setting condition]

- Writing 1 to one of these bits.

[Clearing condition]

- SW: Writing 1 to one of CAEID1.PWMFLOD bits.

31.3.4.6 CAEID1 : Error Interrupt Disable Register 1

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0418

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	PWMFLOD2 to PWMFLOD0		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	PWMCLOD2 to PWMCLOD0		
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
2:0	PWMCLOD2 to PWMCLOD0	Port n Watermark Critical Level Overtok Disable (n = 0 to 2)	R
15:3	—	These bits are read as 0. The write value should be 0.	R
18:16	PWMFLOD2 to PWMFLOD0	Port n Watermark Flush Level Overtok Disable (n = 0 to 2)	R
31:19	—	These bits are read as 0. The write value should be 0.	R

PWMCLODn bits (Port n Watermark Critical Level Overtok Disable (n = 0 to 2))

[Clearing condition]

- SW: Writing 1 to one of these bits clears the corresponding CAEIE1.PWMCLOE bit.

PWMFLODn bits (Port n Watermark Flush Level Overtok Disable (n = 0 to 2))

[Clearing condition]

- SW: Writing 1 to one of these bits clears the corresponding CAEIE1.PWMFLOE bit.

31.3.4.7 CAMIS0 : Monitoring Interrupt Status Register 0

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0440

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PFS1 to PFS0	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	PFS1 to PFS0	Pause Frame Status Flag n (n= 0 to 1)	R/W ^{*1}
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

PFSn bits (Pause Frame Status Flag n (n= 0 to 1))

[Setting condition]

- HW: For bit i, when pause frame i begins (pause assertion timing), see [section 31.5.2. Pause Function](#).

[Clearing condition]

- SW: Writing 1 to one of these bits.

31.3.4.8 CAMIE0 : Monitoring Interrupt Enable Register 0

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0444

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PFE1 to PFE0	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	PFE1 to PFE0	Pause Frame Enable n (n = 0 to 1)	R/W ^{*1}
31:2	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

PFE n bits (Pause Frame Enable n (n = 0 to 1))

[Setting condition]

- Writing 1 to one of these bits.

[Clearing condition]

- SW: Writing 1 to one of CAMID0.PFD bits.

31.3.4.9 CAMID0 : Monitoring Interrupt Disable Register 0

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0448

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PFD1 to PFD0	
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	PFD1 to PFD0	Pause Frame Disable n (n = 0 to 1)	R
31:2	—	These bits are read as 0. The write value should be 0.	R

PFDn bits (Pause Frame Disable n (n = 0 to 1))

[Clearing condition]

- SW: Writing 1 to one of these bits clears the corresponding CAMIE0.PFE bit.

31.3.4.10 CAMIS1 : Monitoring Interrupt Status Register 1

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0450

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	PPFS2 1	PPFS2 0	PPFS1 1	PPFS1 0	PPFS0 1	PPFS0 0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PPFS00	Port 0 Pause Frame Status Flag 0	R/W ^{*1}
1	PPFS01	Port 0 Pause Frame Status Flag 1	R/W ^{*1}
2	PPFS10	Port 1 Pause Frame Status Flag 0	R/W ^{*1}
3	PPFS11	Port 1 Pause Frame Status Flag 1	R/W ^{*1}
4	PPFS20	Port 2 Pause Frame Status Flag 0	R/W ^{*1}
5	PPFS21	Port 2 Pause Frame Status Flag 1	R/W ^{*1}
31:6	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

PPFSmn bits (Port m Pause Frame Status n)

[Setting condition]

- HW: For bit i, when pause frame i begins (pause assertion timing), see [section 31.5.2. Pause Function](#).

[Clearing condition]

- SW: Writing 1 to one of these bits.

31.3.4.11 CAMIE1 : Monitoring Interrupt Enable Register 1

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0454

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	PPFE2 1	PPFE2 0	PPFE1 1	PPFE1 0	PPFE0 1	PPFE0 0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PPFE00	Port 0 Pause Frame Enable 0	R/W ^{*1}
1	PPFE01	Port 0 Pause Frame Enable 1	R/W ^{*1}
2	PPFE10	Port 1 Pause Frame Enable 0	R/W ^{*1}
3	PPFE11	Port 1 Pause Frame Enable 1	R/W ^{*1}
4	PPFE20	Port 2 Pause Frame Enable 0	R/W ^{*1}
5	PPFE21	Port 2 Pause Frame Enable 1	R/W ^{*1}
31:6	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

PPFEmn bits (Port m Pause Frame Enable n)

[Setting condition]

- Writing 1 to one of these bits.

[Clearing condition]

- SW: Writing 1 to one of CAMID1.PPFD bits.

31.3.4.12 CAMID1 : Monitoring Interrupt Disable Register 1

Base address: COMA = 0x403C_9000
COMA_NS = 0x503C_9000

Offset address: 0x0458

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	PPFD 21	PPFD 20	PPFD 11	PPFD 10	PPFD 01	PPFD 00
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	PPFD00	Port 0 Pause Frame Disable 0	W ^{*1}
1	PPFD01	Port 0 Pause Frame Disable 1	W ^{*1}
2	PPFD10	Port 1 Pause Frame Disable 0	W ^{*1}
3	PPFD11	Port 1 Pause Frame Disable 1	W ^{*1}
4	PPFD20	Port 2 Pause Frame Disable 0	W ^{*1}
5	PPFD21	Port 2 Pause Frame Disable 1	W ^{*1}
31:6	—	These bits are read as 0. The write value should be 0.	R/W

Note 1. Read value differs from written value.

PPFDmn bits (Port m Pause Frame Disable n)

[Clearing condition]

- SW: Writing 1 to one of these bits clears the corresponding CAMIE1.PPFE bit.

31.4 Register Utilization

31.4.1 Software Flows

SW: Follow the flows described in this section.

31.4.1.1 Software Flow Legend

Figure 31.2 shows the software flow legend.

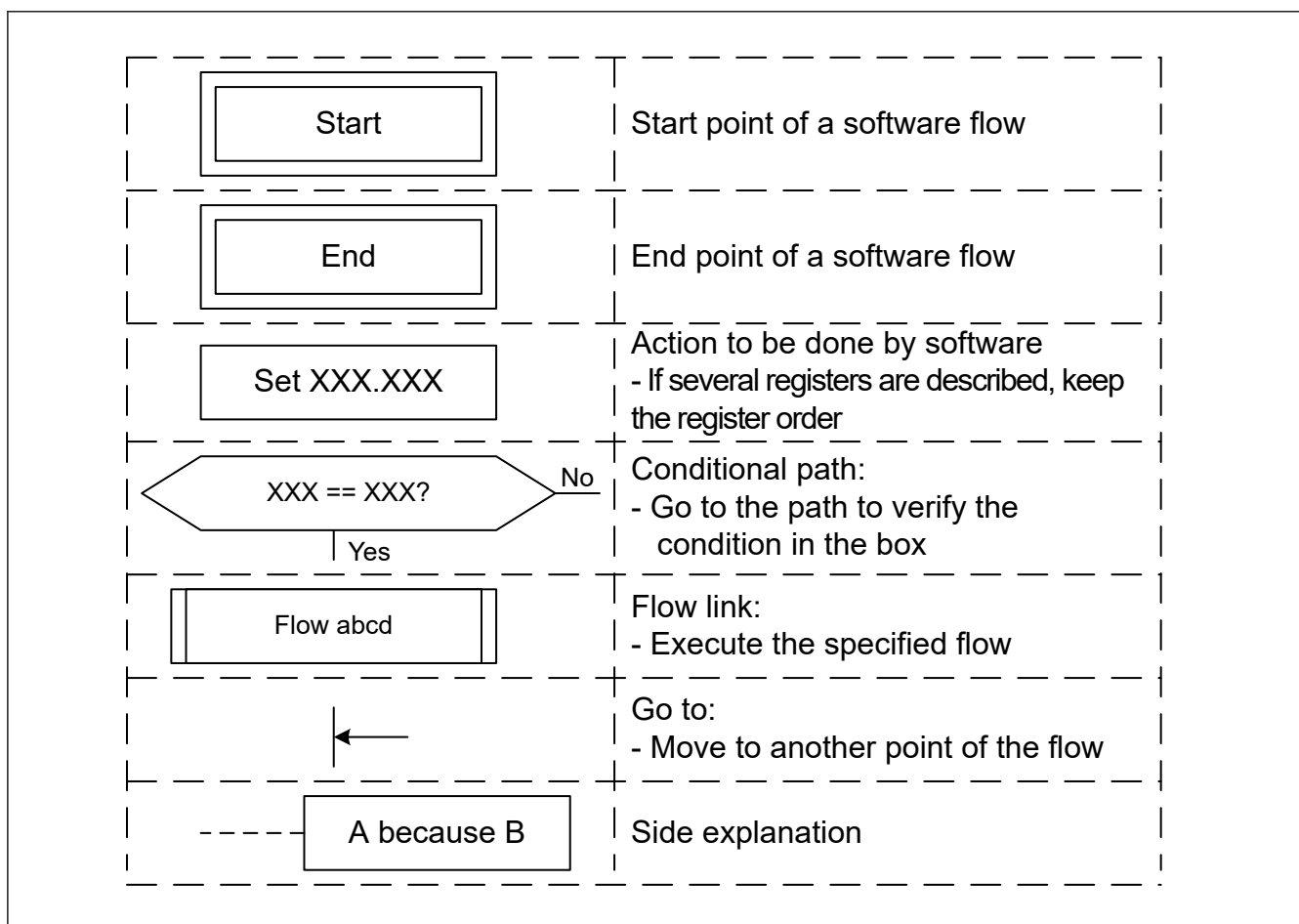


Figure 31.2 Software flow legend

31.4.1.2 Interrupt Handling Flow

Interrupt handling flow can be used in any mode except Reset mode.

Figure 31.3 shows the interrupt handling flow.

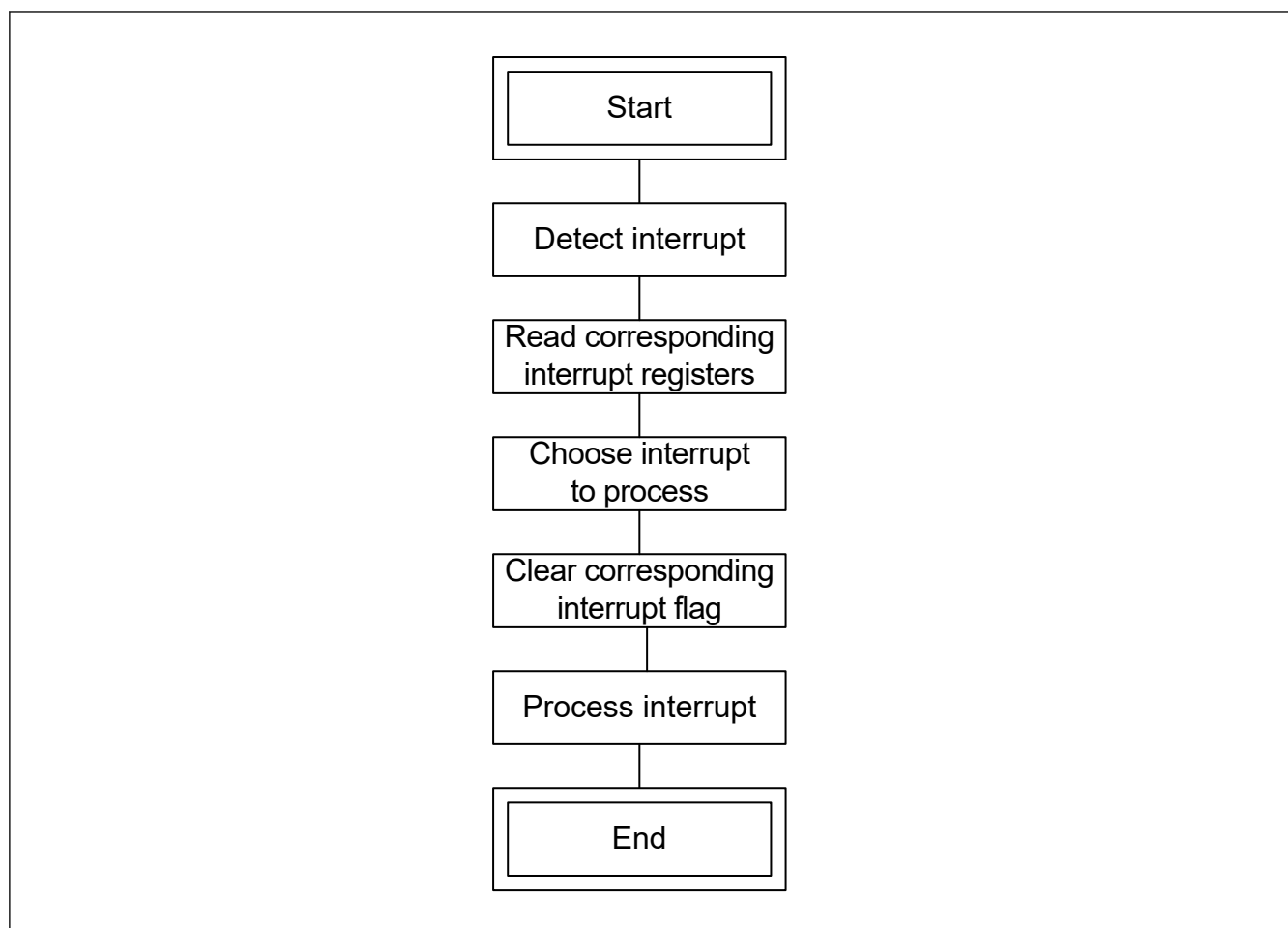


Figure 31.3 Interrupt handling flow

31.4.1.3 Called Software Flows

The flows described in this section can only be called from other flows as indicated in the flow link step in [Figure 31.2](#) and cannot be used alone.

31.4.1.3.1 Buffer Pool Initialization Flow

[Figure 31.4](#) shows the buffer pool initialization flow.

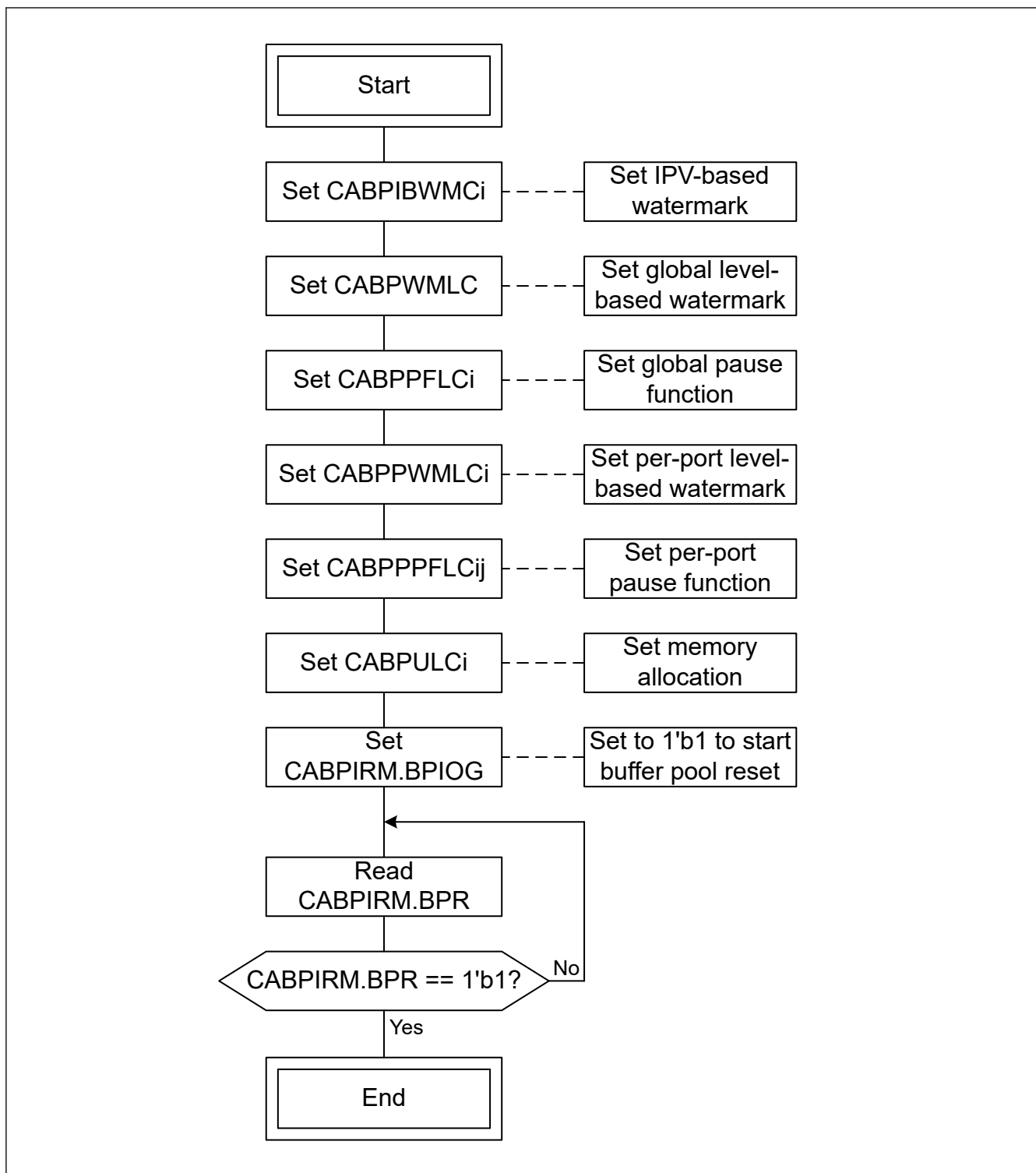


Figure 31.4 Buffer pool initialization flow

31.4.1.3.2 ESWM Reset Flow

Figure 31.5 shows the ESWM reset flow.

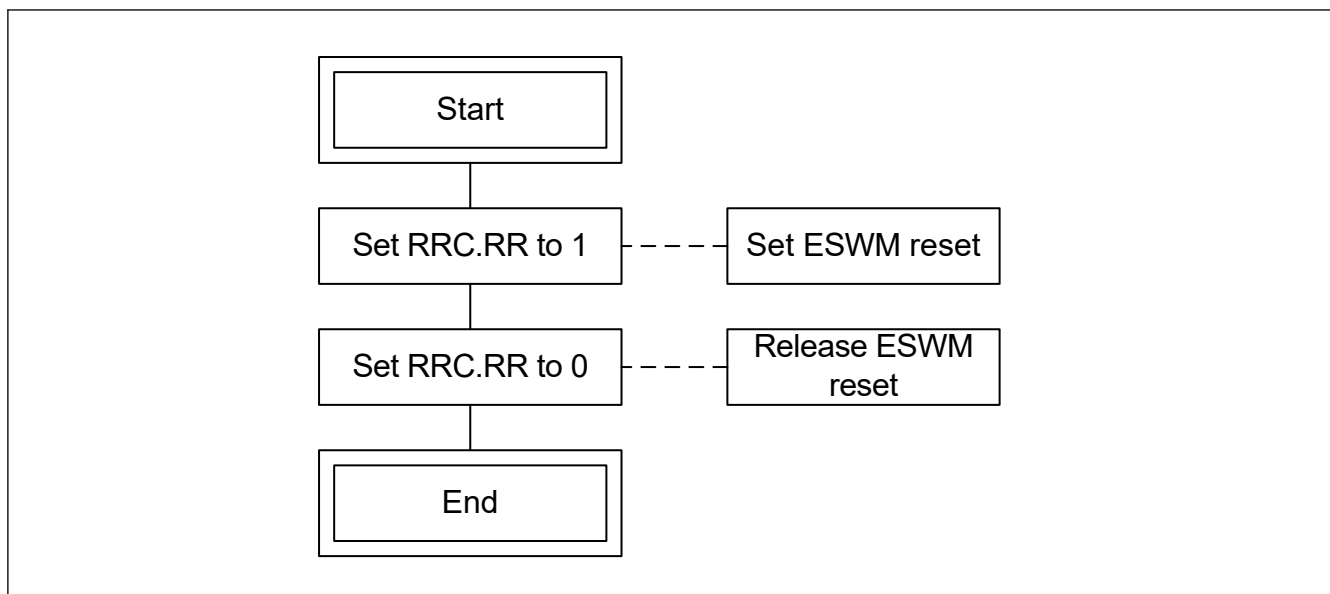


Figure 31.5 ESWM reset flow

31.4.1.3.3 Agent i Clock Control Flows (i = 0 to 3)

Figure 31.6, Figure 31.7 and Figure 31.8 show the agent i clock control flows.

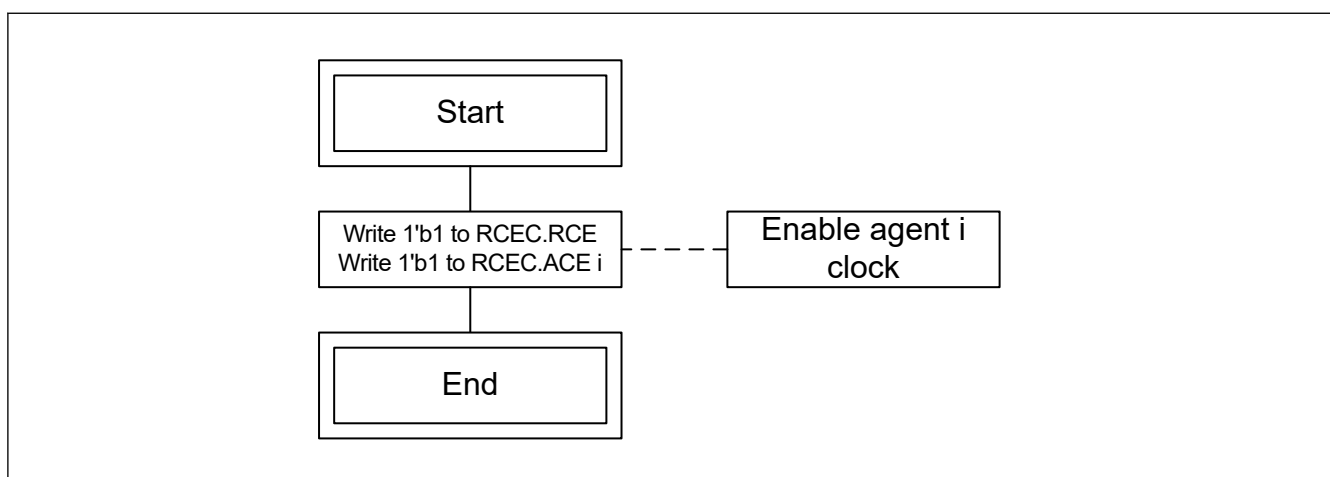


Figure 31.6 Agent i clock enable flow

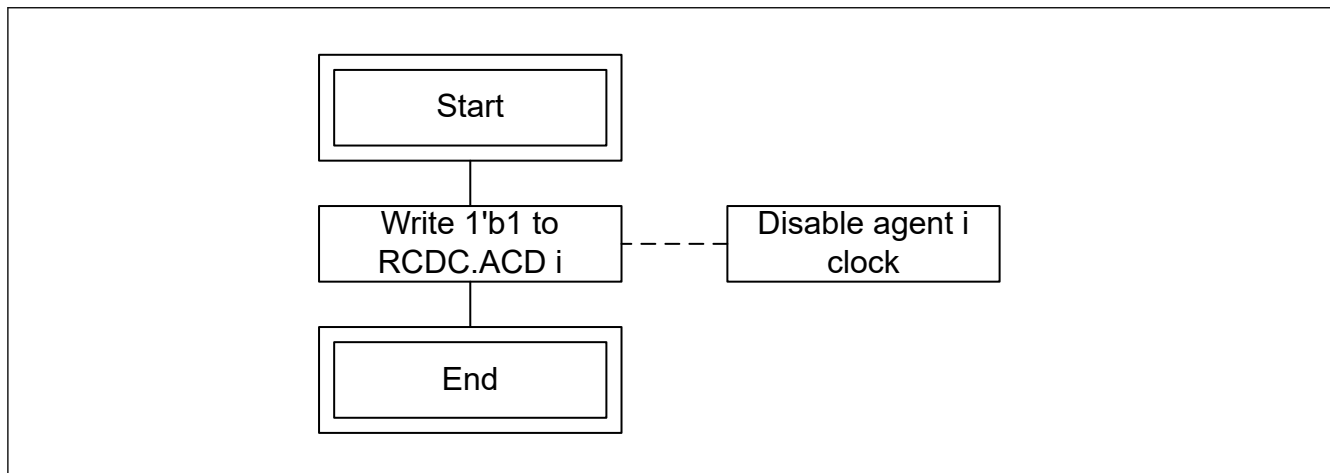


Figure 31.7 Agent i clock disable flow

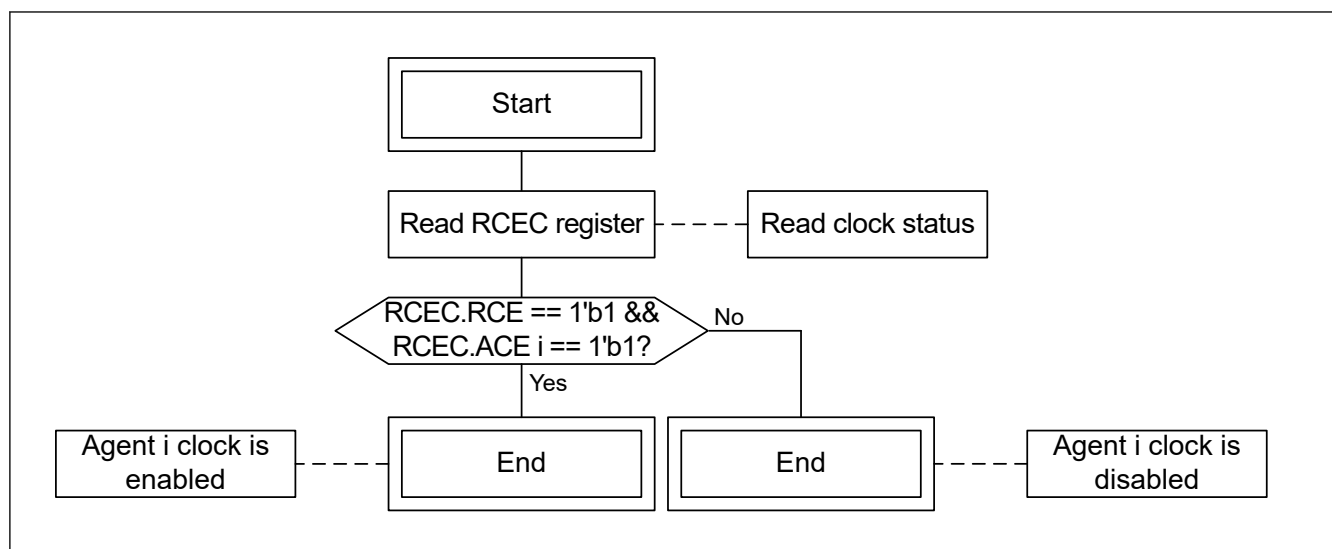


Figure 31.8 Agent i clock status check flow

31.4.1.3.4 Switch Clock Control Flows

Figure 31.9 and Figure 31.10 show the switch clock control flows.

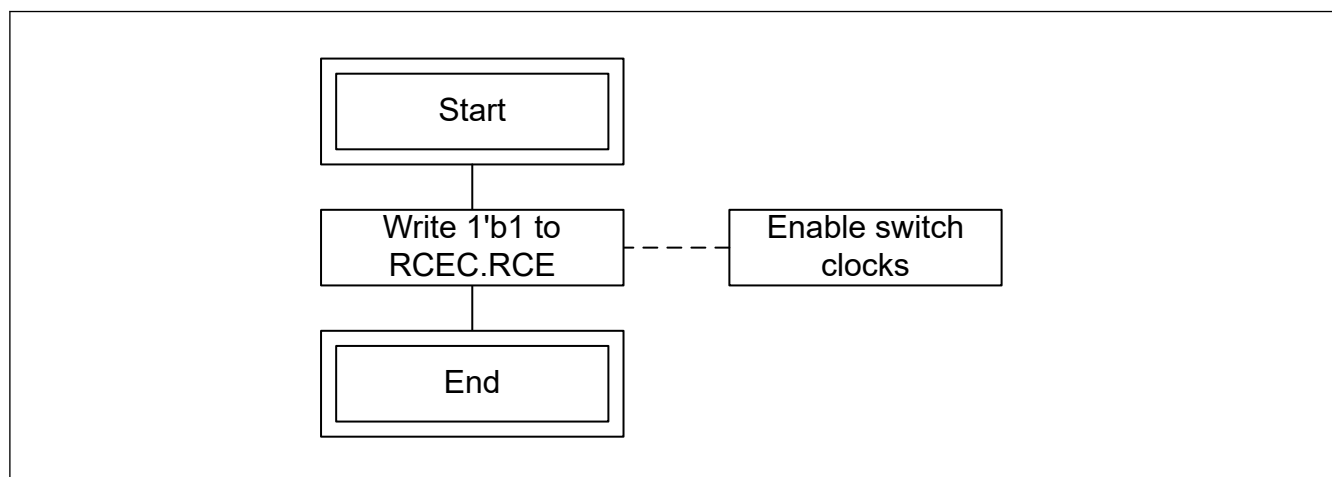


Figure 31.9 Switch clock enable flow

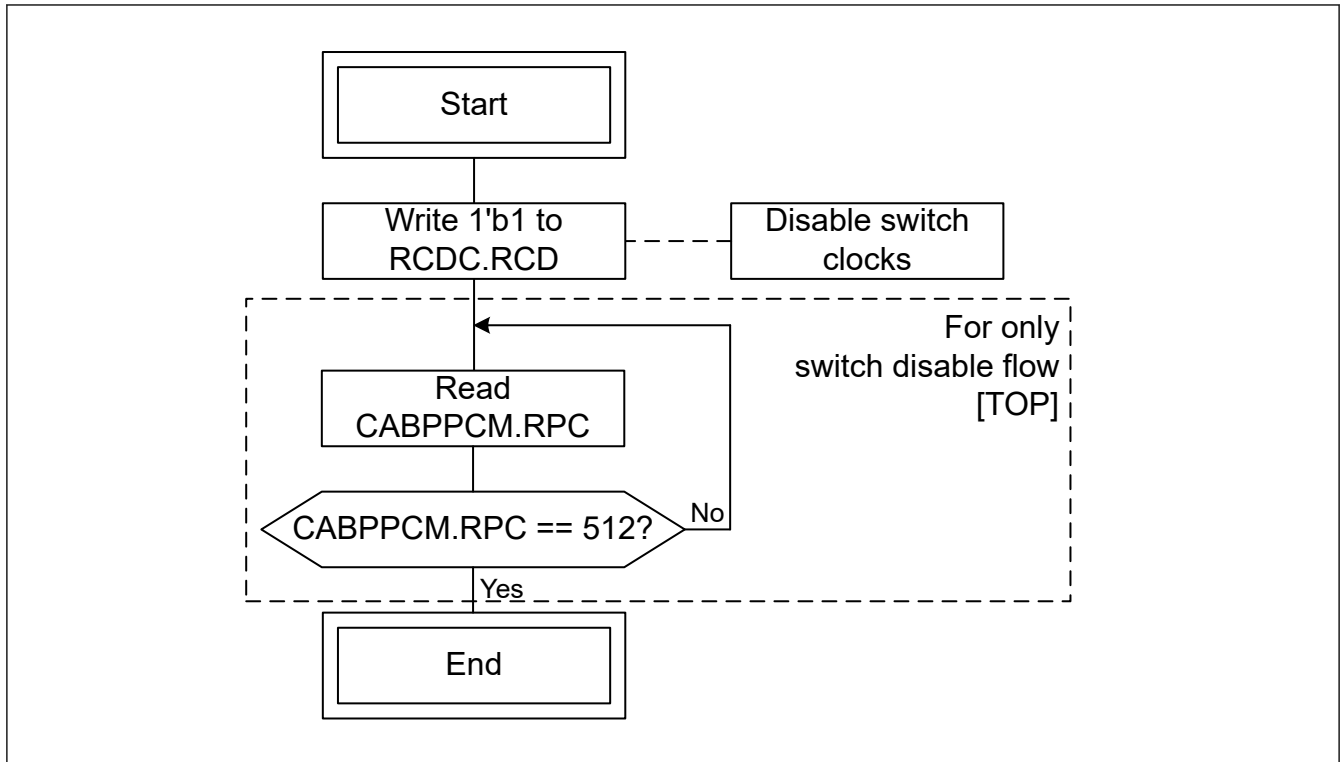


Figure 31.10 Switch clock disable flow

31.4.1.4 Register Writable Without Software Flow

This section describes registers that have not been described. These registers can be changed dynamically. However, it is necessary that the initial settings such as clock enabling have been completed.

31.5 Functional Description

31.5.1 Watermark Function

If a switch overflow occurs, frames in the switch are lost at the ingress part of agents [GWCA] [ETHA]. This result in a random loss of data is independent of the actual priority of frames. To avoid this, the watermark function allows to selectively reject frames in MFWD before a switch overflow occurs.

Ethernet Common Agent includes three watermark functions:

- IPV-based watermark
- Global level-based watermark
- Per-port level-based watermark

The watermark functions aim at controlling the switch behavior in case of overflow by generating data loss. All watermark functions are running in parallel and a frame can be rejected by one or more watermark functions.

31.5.1.1 IPV-based Watermark

The IPV-based watermark ensures the minimum QoS requirement for a switch. It is enabled by the default configurations for a switch maximum frame size of 256 bytes. This function is controlled by the Ethernet Common Agent, however, data loss is performed by the Forwarding Engine [FWD]. IPV-based watermark with a combination of other functions, is affected by the queue pause regardless of frame size (256).

[Figure 31.11](#) shows an example of IPV-based unsecure watermark setting for Pointer number to address local RAM == 4096. The IPV-based secure watermark works the same way but using CABPIBWMCi.IBSWMPN registers.

Note: The variations of 512 - CABPPCM.RPC in [Figure 31.11](#) are arbitrary and do not correspond to anything relevant.

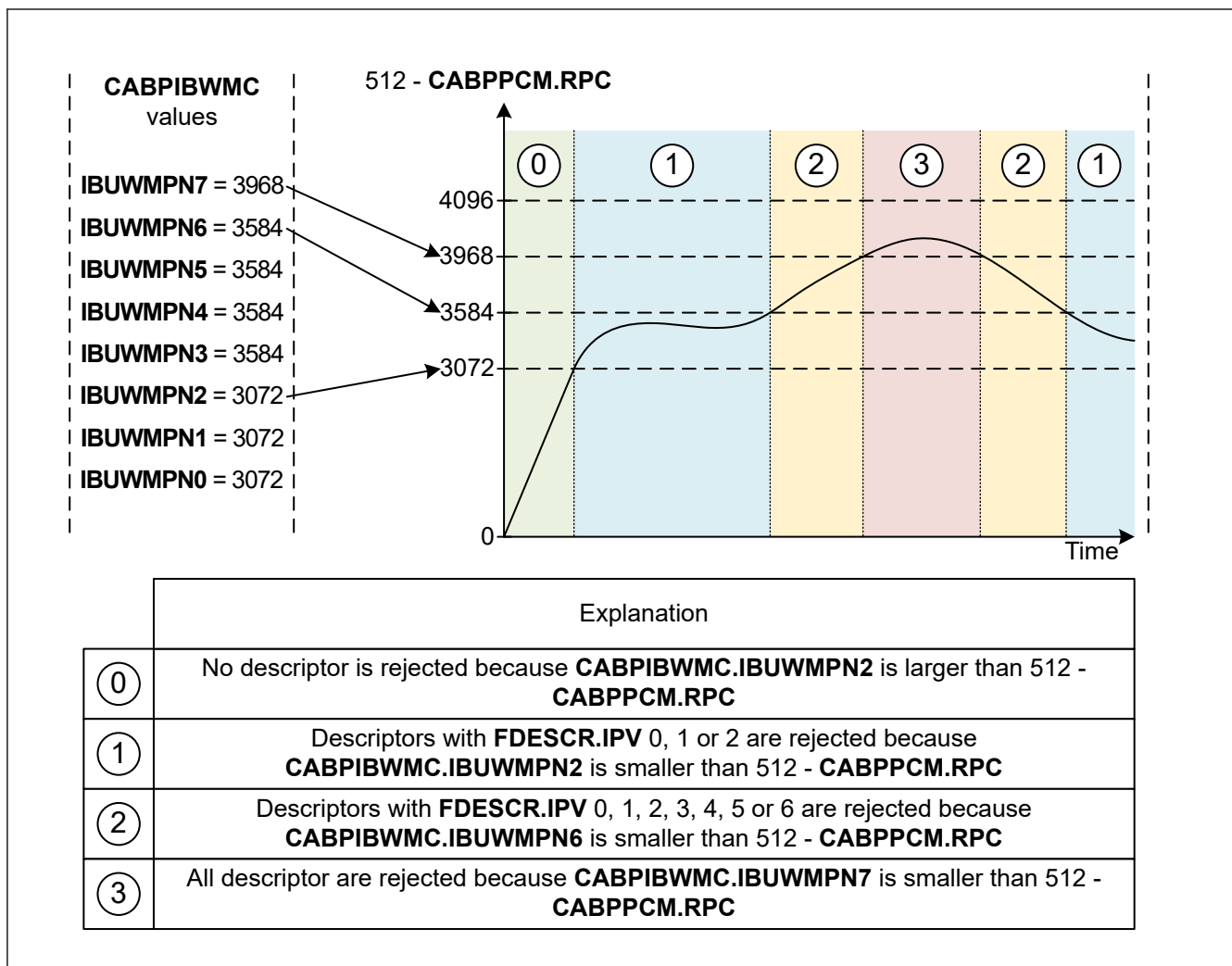


Figure 31.11 Example for IPV-based watermark setting

Note: FDESCR.IPV refers to the internal priority value given to a frame by forwarding engine in the forwarding descriptor but before filtering. For more details, see the Forwarding Engine specification [FWD].

31.5.1.2 Global Level-based Watermark

The global level-based watermark aims at rejecting frames with targeted IPV's depending on the number of pointers currently used by the switch (CABPPCM.RPC register). This function is controlled by the Ethernet Common Agent (through WM.CRITICAL[3:0] and WM.FLUSH[3:0]), however, data loss is performed by the Forwarding Engine [FWD].

Note: WM.CRITICAL and WM.FLUSH interfaces can also be set by the per-port level-based watermark (one of these pins is set if the per-port level-based watermark or the global level-based watermark is in the required setting conditions).

Figure 31.12 shows an example for global level-based watermark setting for Pointer number to address local RAM = 4096.

Note: The variations of 512 - CABPPCM.RPC in Figure 31.12 are arbitrary and do not correspond to anything relevant.

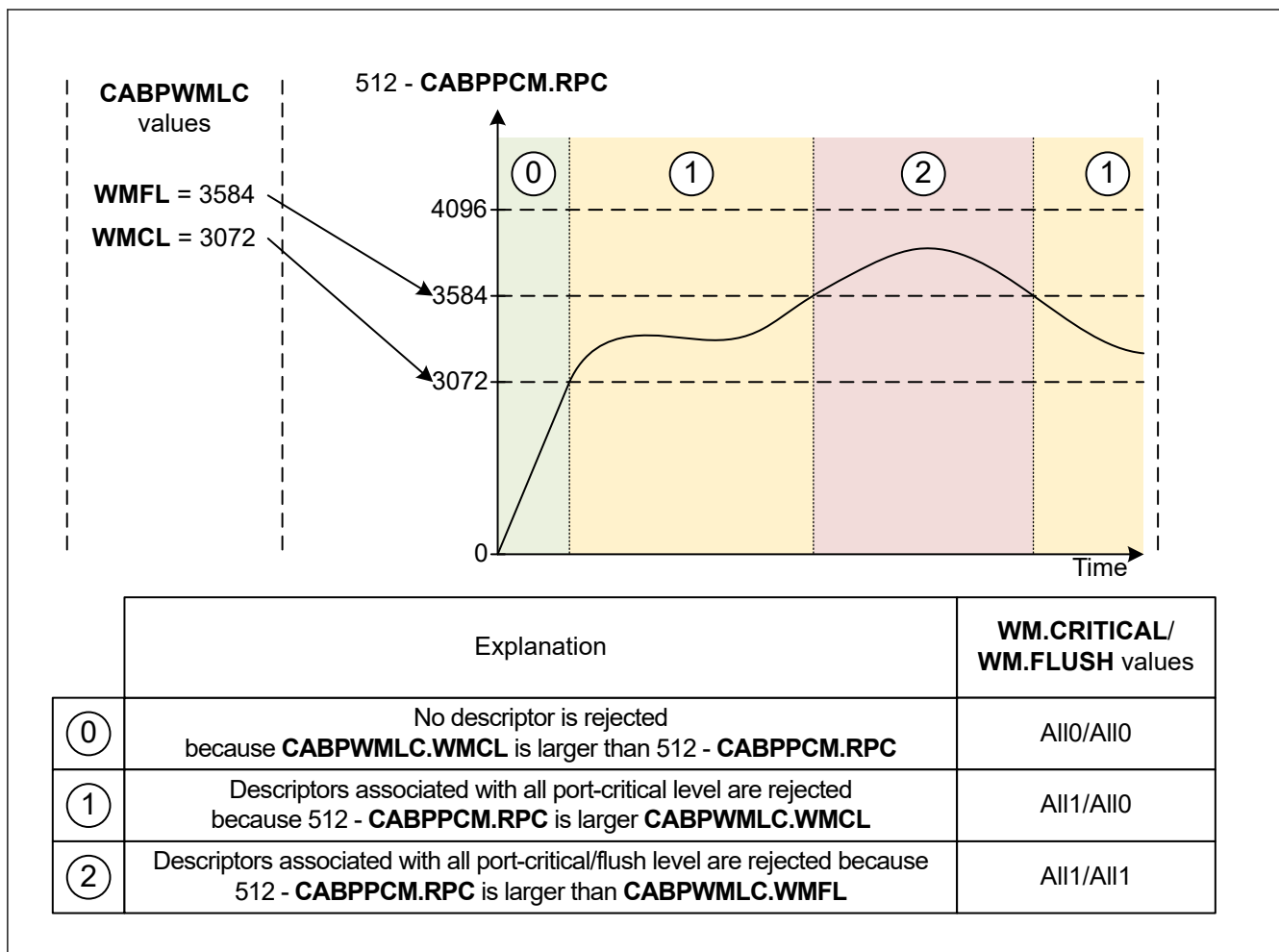


Figure 31.12 Example for global level-based watermark setting

31.5.1.3 Per-port Level-based Watermark i (i = 0 to 3)

The per-port level-based watermark aims at rejecting frames with targeted IPv4s depending on **CABPPWMLCi.PWMCL**. This function is controlled by the Ethernet Common Agent (through **WM.CRITICAL[i]** and **WM.FLUSH[i]**), however, data loss is performed by the Forwarding Engine [FWD].

Note: **WM.CRITICAL** and **WM.FLUSH** interfaces can also be set by the global level-based watermark (one of these pins is set if the per-port level-based watermark or the global level-based watermark is in the required setting conditions).

[Figure 31.13](#) shows an example of per-port level-based watermark setting for port i when Pointer number to address local RAM = 4096.

Note: The variations of **CABPCPMi.RPCP** in [Figure 31.13](#) are arbitrary and do not correspond to anything relevant.

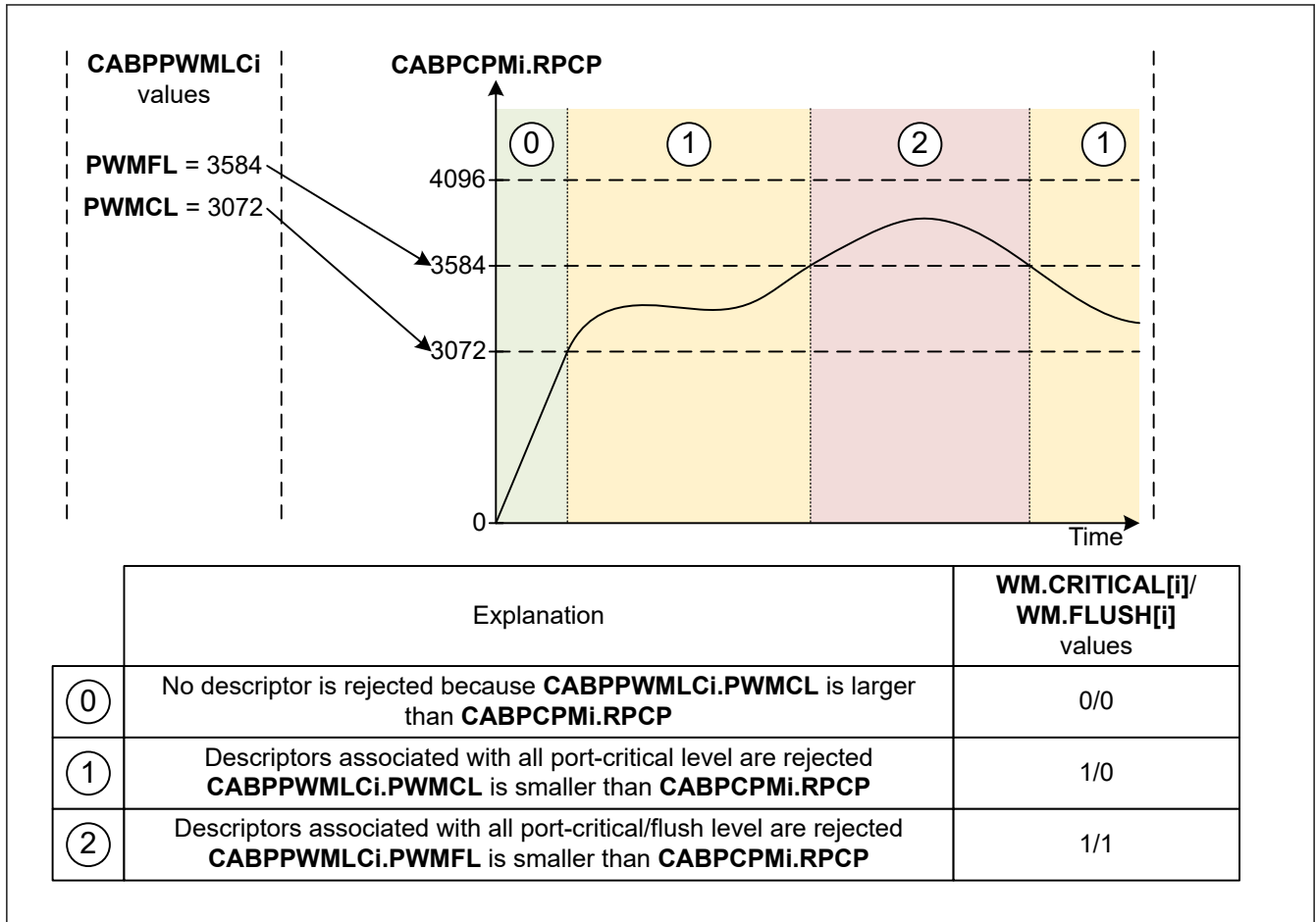


Figure 31.13 Example for per-port level-based watermark setting for port i

31.5.2 Pause Function

If a switch overflow occurs, frames in the switch are lost at the ingress part of agents [GWCA] [ETHA]. This result in a random loss of data is independent of the actual priority of frames. To avoid this, the pause function allows to selectively pause frame injection by agents [GWCA] [ETHA].

Ethernet Common Agent includes two pause functions, the global pause function and the per-port pause function. The pause functions aim at controlling the switch behavior in case of overflow by generating backpressure. All pause functions are running in parallel and backpressure can be applied by one or more pause functions.

31.5.2.1 Global Pause Function

The global pause function aims at applying backpressure on agents for targeted IPVs depending on CABPPCM.RPC. The pause function can pause agents with two combinations of IPVs.

The global pause function is controlled by the Ethernet Common Agent PAUSE[3:0][1:0] interface, however, backpressure is performed by the agents [ETHA] [GWCA].

Note: The pause interface can also be set by the per-port pause function (one of these pins is set if the per-port pause function or the global pause function is in the required setting conditions).

Figure 31.14 shows an example of global pause function setting for Pause level number = 2 and Pointer number to address local RAM = 4096.

Note: The variations of 512 - CABPPCM.RPC in Figure 31.14 are arbitrary and do not correspond to anything relevant.

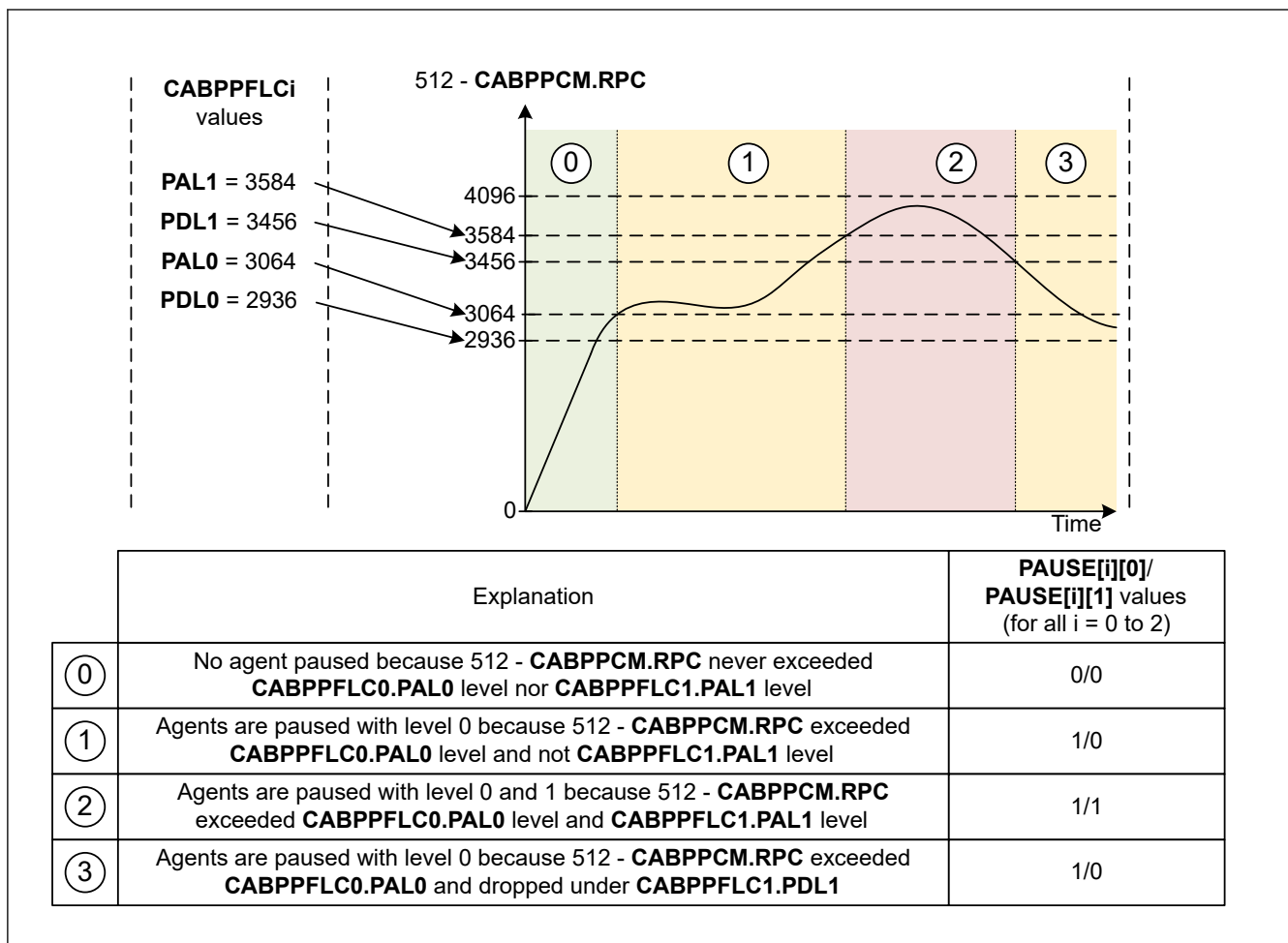


Figure 31.14 Example for global pause function setting

31.5.2.2 Per-port Pause Function i (i = 0 to 3)

The per-port pause function aims at applying backpressure on agents for targeted IPv4s depending on CABPPWMLCi.PWMCL. The pause function can pause agents with two combinations of IPv4s.

This function is controlled by the Ethernet Common Agent PAUSE[i] interface, however, backpressure is performed by the agents [ETHA] [GWCA].

Note: The pause interface can also be set by the per-port pause function (one of these pins is set if the per-port pause function or the global pause function is in the required setting conditions).

Figure 31.15 shows an example of per-port pause function setting for port i when Pause level number = 2 and Pointer number to address local RAM = 4096.

Note: The variations of CABPPWMLCi.PWMCL in Figure 31.15 are arbitrary and do not correspond to anything relevant.

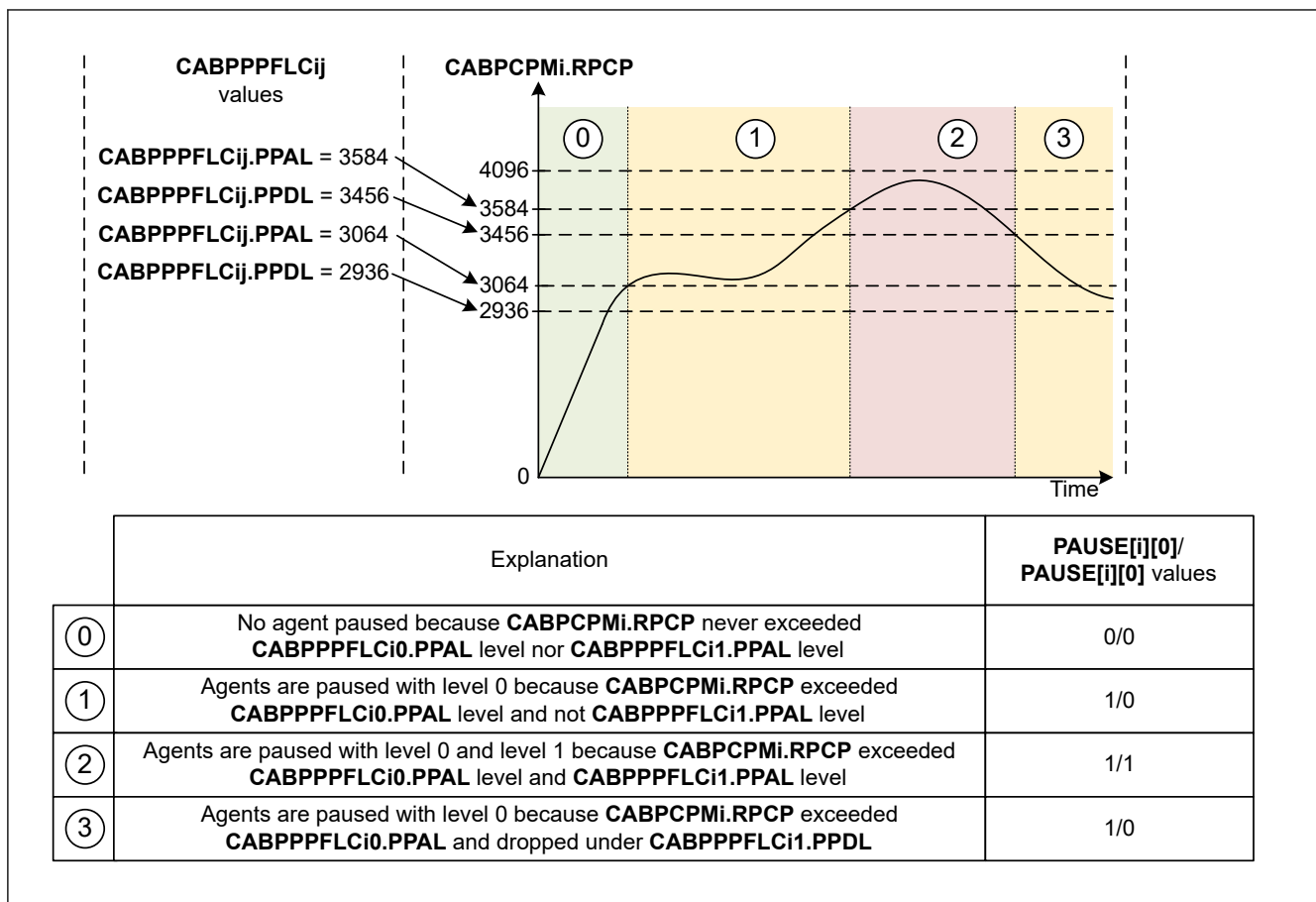


Figure 31.15 Example for per-port pause function setting for port i

31.5.3 Per-port Memory Allocation Function

It is possible to control the maximum and the minimum number of pointers that can be used per port due to CABPULCi.MNNPN and CABPULCi.MXNPN. Any agent can use at least its minimum number of pointers and at most its maximum number of pointers. This section provides some setting examples for the memory allocation function.

31.5.3.1 Shared Memory Setting (Default Setting)

In shared memory, the data RAM is shared by all the ports and is the reset state of the Ethernet Common Agent.

Figure 31.16 describes the shared memory setting for Pointer number to address local RAM = 4096 and Port number on the switch = 3.

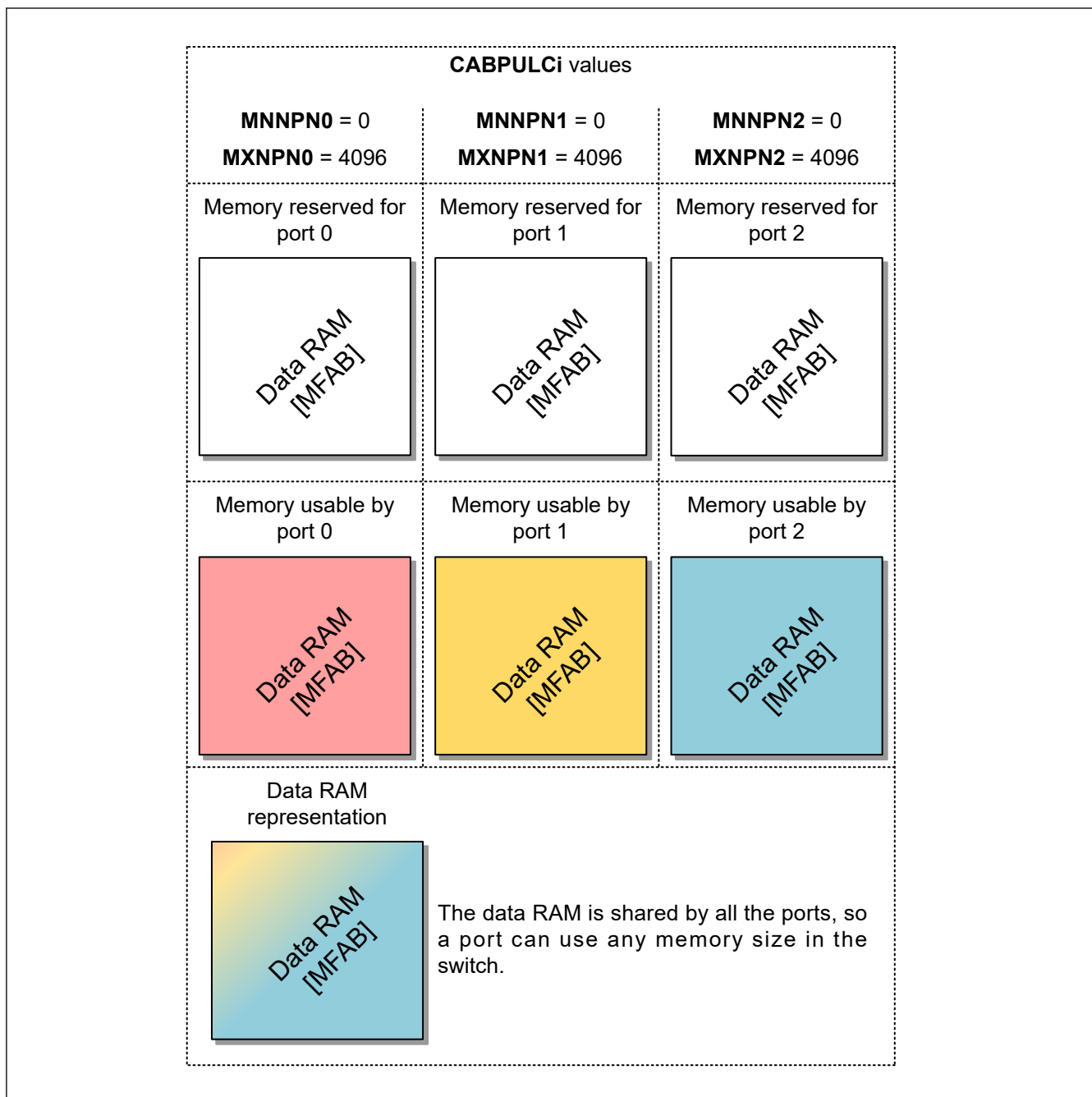


Figure 31.16 Shared memory setting

31.5.3.2 Reduced Memory Setting 1 (Recommended Setting)

In reduced memory, the data RAM is shared by all agents but some of them are not able to use all the memory because they can make the switch overflow (especially for CPU ports [GWCA]). To set this mode, use the CABPULCi.MXNPN register.

Figure 31.17 shows an example of reduced memory setting for Pointer number to address local RAM = 4096 and Port number on the switch = 3. For example, if port 2 is a fast CPU port, CABPULCi.MXNPN register can limit the number of reception frames.

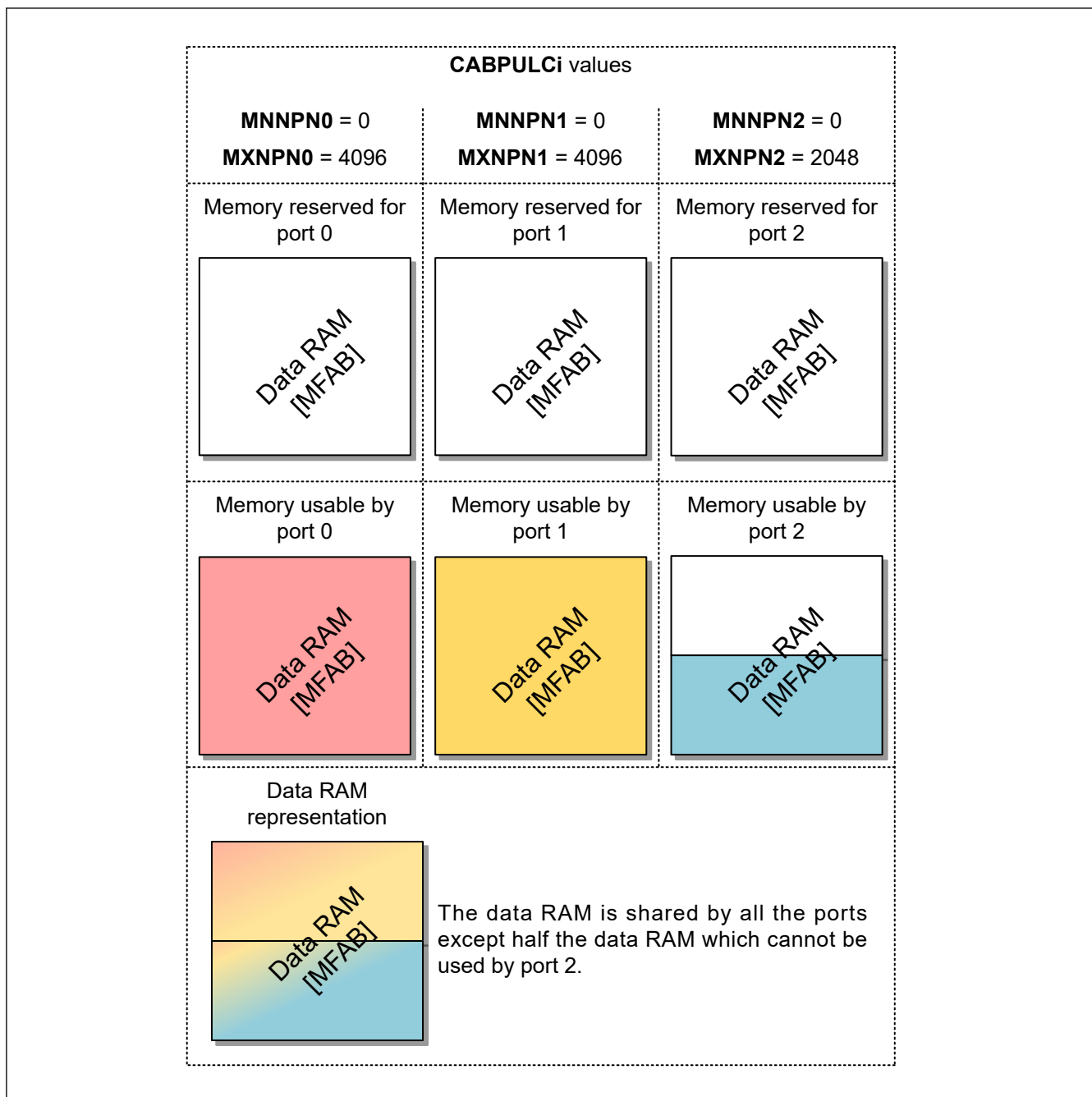


Figure 31.17 Reduced memory setting example

31.5.3.3 Reduced Memory Setting 2 (Recommended Setting)

In reduced memory, the data RAM is shared by all agents but some of them are not able to use all the memory because they can make the switch overflow (especially for CPU ports [GWCA]). To set this mode, use the CABPULCi.MNNPN register.

Figure 31.18 shows an example of reduced memory setting for Pointer number to address local RAM = 4096 and Port number on the switch = 3. For example, if port 0 or port 1 is a slow Ethernet port, CABPULCi.MNNPN register can protect the number of frame buffer.

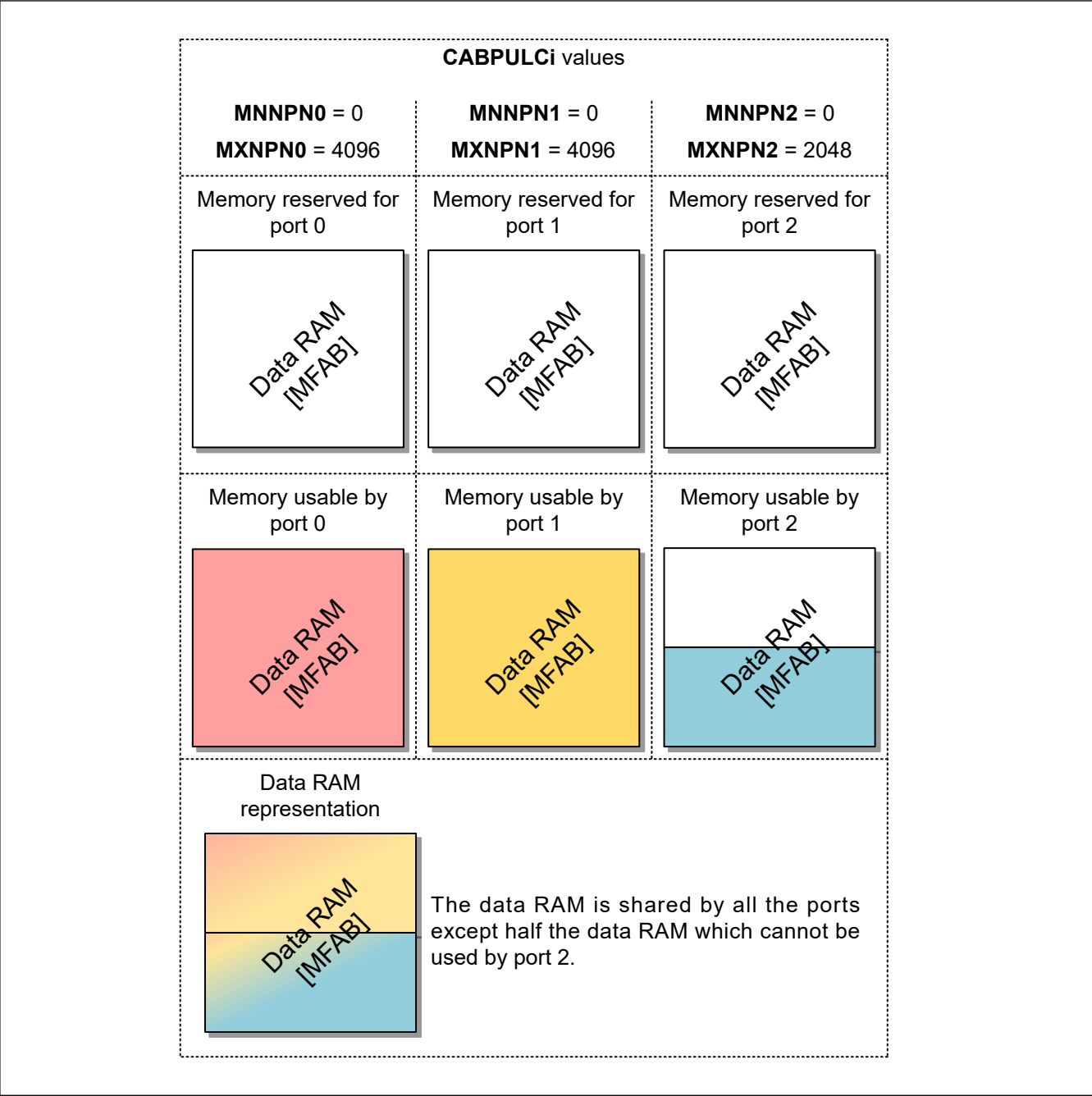


Figure 31.18 Reduced memory setting example

31.5.3.4 Hybrid Memory Setting

All the setting described in [section 31.5.3.1. Shared Memory Setting \(Default Setting\)](#), [section 31.5.3.2. Reduced Memory Setting 1 \(Recommended Setting\)](#) and [section 31.5.3.3. Reduced Memory Setting 2 \(Recommended Setting\)](#) can be used in combination to describe a complex memory such as the hybrid memory setting.

[Figure 31.19](#) shows an example of hybrid memory setting for Pointer number to address local RAM = = 4096 and Port number on the switch = = 4.

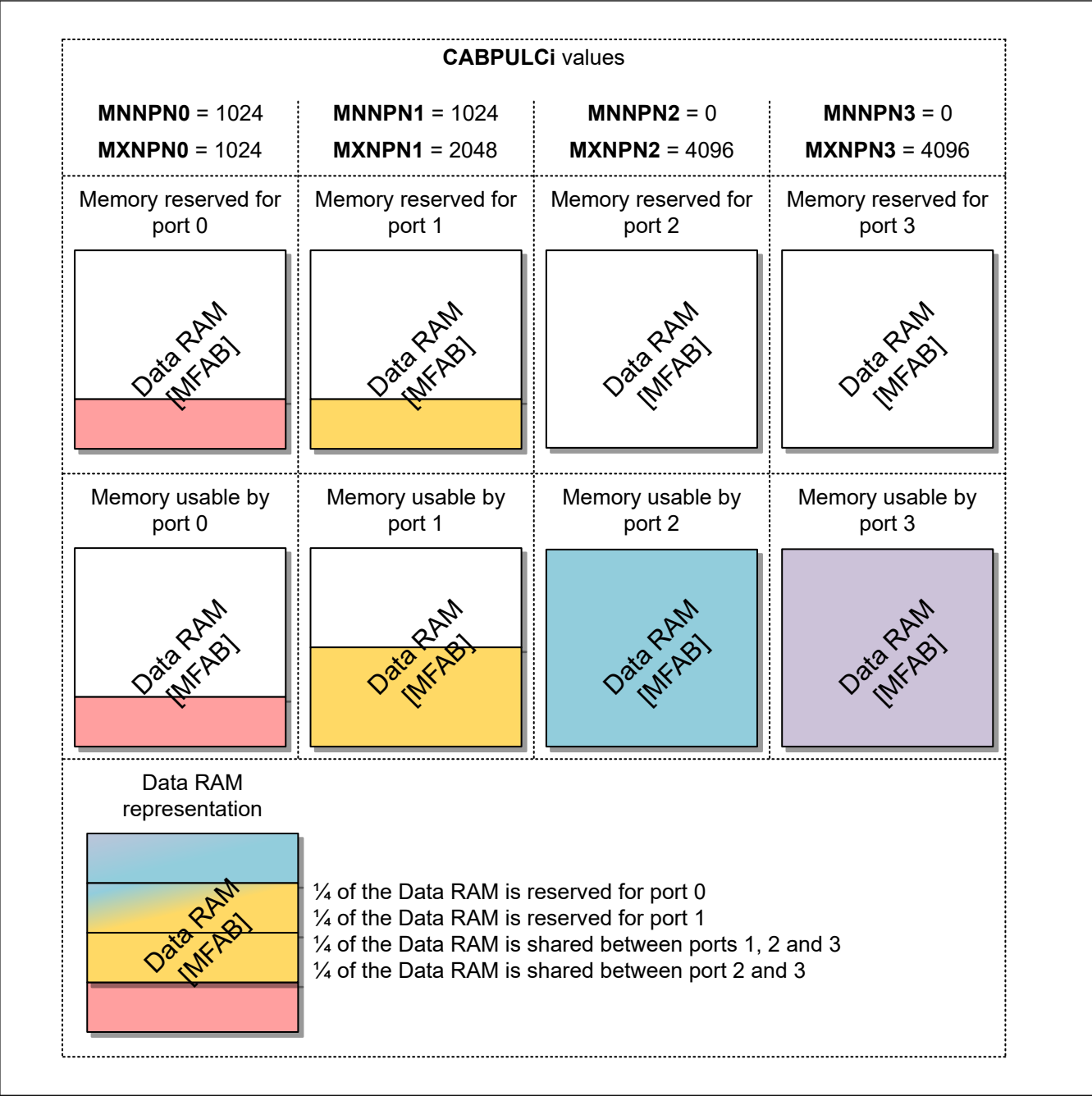


Figure 31.19 Hybrid memory setting example

31.6 Precautions

31.6.1 Precautions

Not applicable.

31.6.2 Restrictions (Including Known Problems)

Not applicable.